

HW1: Introduction to Cryptography

CYB-402

Tyler Vander Mooren

1.1. - No.

1.2.

Since the key is unknown, Anthony now must try all possible shifts. This results in the ciphertext being decrypted to valid locations at different shifts. Which are 4 = Arena, and 13 = river. Because of this Anthony cannot uniquely determine the meeting location.

2.1

128^8

2.2

56 bits

2.3

37.6 bits, 38 bits

2.4.1

19 characters

2.4.2

27 characters

3.1

$6 \bmod 13$

3.2

$6 \bmod 13$

3.3

$6 \bmod 13$

3.4

$6 \bmod 13$

3.cont. In all four problems they give the same result, because the numbers belong to the same equivalence class they will behave the same way under multiplication. So in this problems all the numbers reduce to $2 \cdot 3 \bmod 13$, giving the result $6 \mid 13 \mid$.

4.1. $1/5 \bmod 13$

$$5 \cdot 1 = 5 \neq 1$$

$$5 \cdot 2 = 10 \neq 1$$

$$5 \cdot 3 = 15 \equiv 2 \bmod 13$$

$$5 \cdot 5 = 25 \equiv 12 \bmod 13$$

$$5 \cdot 8 = 40$$

$$= 39 + 1$$

$$= (13 \cdot 3) + 1$$

$$= 1 \bmod 13$$

Answer: $1/5 \bmod 13 = 8$

4.2. $1/5 \bmod 7$

$$5 \cdot 1 = 5 \neq 1$$

$$5 \cdot 2 = 10 \equiv 3 \neq 1$$

$$5 \cdot 3 = 15 \equiv 1 \bmod 7$$

Answer: $1/5 \bmod 7 = 3$

4.3. $3 \cdot 2/5 \bmod 7$

$$= 3 \cdot 2 \cdot 5^{-1} \bmod 7$$

From 4.2 we get $5^{-1} \equiv 3 \bmod 7$.

$$= 3 \cdot 2 \cdot 3 \bmod 7$$

Answer = $18 \bmod 7 = 4$

5.

	0	1	2	3
0	0	1	2	3

1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

6.

	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

7.

	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	2	3

3	0	3	1	4	2
4	0	4	3	2	1

8.

	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

9.

Z_{11}, elements = {0,1,2,..., 10} $5 \cdot x \equiv 1 \pmod{11}$	Z_{12}, elements = {0,1,2,..., 11} $5 \cdot x \equiv 1 \pmod{12}$	Z_{13}, elements = {0,1,2,..., 12} $5 \cdot x \equiv 1 \pmod{13}$
$5 \times 1 \pmod{11} = 5$	$5 \times 3 \pmod{11} = 5$	$5 \times 3 \pmod{11} = 5$
$5 \times 2 \pmod{11} = 10$	$5 \times 3 \pmod{11} = 10$	$5 \times 3 \pmod{11} = 10$

$5 \times 3 \bmod 11 = 4$	$5 \times 3 \bmod 11 = 3$	$5 \times 3 \bmod 11 = 2$
$5 \times 3 \bmod 11 = 9$	$5 \times 3 \bmod 11 = 8$	$5 \times 3 \bmod 11 = 7$
$5 \times 3 \bmod 11 = 3$	$5 \times 3 \bmod 11 = 1$	$5 \times 3 \bmod 11 = 12$
$5 \times 3 \bmod 11 = 8$	$5 \times 3 \bmod 11 = 6$	$5 \times 3 \bmod 11 = 4$
$5 \times 3 \bmod 11 = 2$	$5 \times 3 \bmod 11 = 11$	$5 \times 3 \bmod 11 = 9$
$5 \times 3 \bmod 11 = 7$	$5 \times 3 \bmod 11 = 4$	$5 \times 3 \bmod 11 = 1$
$5 \times 3 \bmod 11 = 1$	$5 \times 3 \bmod 11 = 9$	$5 \times 3 \bmod 11 = 6$
$5 \times 3 \bmod 11 = 6$	$5 \times 3 \bmod 11 = 2$	$5 \times 3 \bmod 11 = 11$
	$5 \times 3 \bmod 11 = 7$	$5 \times 3 \bmod 11 = 3$
		$5 \times 3 \bmod 11 = 8$
$5 \times 9 = 45$ and $45 \bmod 11 = 1$. Multiplicative inverse of 5 in Z_{11} is 9.	$5 \times 5 = 25$ and $25 \bmod 12 = 1$. Multiplicative inverse of 5 in Z_{12} is 5.	$5 \times 8 = 40$ and $40 \bmod 13 = 1$. Multiplicative inverse of 5 in Z_{13} is 8.

10.1. $x = 9$

10.2. $x = 10$

10.3

$$x = 3^{10} \pmod{13}$$

$$3^{10} = (3^2)^5$$

$$= 9^5 \pmod{13}$$

$$9^2 = 81 \equiv 81 - 78 = 3 \pmod{13}$$

$$9^4 = (9^2)^2 \equiv 3^2 = 9 \pmod{13}$$

$$= 9^4 * 9 \equiv 3^2 * 9 = 81 \equiv \pmod{13}$$

$$9^5 \equiv 3 \pmod{13}$$

$$X = 3$$

10.4

$$X = 7^{100} \pmod{13}$$

$$\gcd(7, 13) = 1$$

$$\gcd(a, 13) = 1, a^{12} \equiv 1 \pmod{13}$$

$$100 \bmod 12 = 4$$

$$100 \equiv 4 \pmod{12}$$

So $7^{100} \equiv 7^4 \pmod{13}$

$$7^2 = 49 \equiv 49 - 39 = 10 \pmod{13}$$

$$7^4 = (7^2)^2 \equiv 10^2 = 100$$

$$100 \pmod{13} = 9, 100 \equiv 9 \pmod{13}$$

So

$$7^{100} \equiv 7^4 \equiv 9 \pmod{13}$$

$$X = 9$$

10.5

k	$K = 1, 2, \dots, 12 ; 7^k \pmod{13}$
1	7
2	$7^2 = 49 \equiv 10$
3	$7^3 = 7^2 * 7 \equiv 10 * 7 = 70 \equiv 5$
4	$7^4 = 5 * 7 = 35 \equiv 9$
5	$7^5 = 9 * 7 = 63 \equiv 1$

$$X = 5$$

$$11.1 \ d(y) = 5(y-2) \pmod{26}$$

11.2 DEG

12.1 MPGXSXPAOGXXAN

$$12.2 \ d(y) = 21(y-6) \pmod{26}$$

$$13. \ E(X) = ax + b \pmod{26}$$

$$l \rightarrow 17 \ h \rightarrow 7 \ a \rightarrow 0 \ o \rightarrow 14$$

$$\text{Need } y \equiv ax + b \pmod{26}$$

$$X = 11 \text{ to } y = 7: 11a + b \equiv 7 \pmod{26}$$

$$X = 0 \text{ to } y = 14: 0 * a + b \equiv 14 \pmod{26} \rightarrow b \equiv 14$$

$$11a \equiv 19 \pmod{26}$$

$$11^{-1}$$

$$26 = 11 * 2 + 4$$

$$11 = 4 * 2 + 3$$

$$4 = 3 * 1 + 1$$

$$1 = 4 - 3 * 1 = 4 - (11 - 4 * 2) = 4 * 3 - 11 \text{ and}$$

$$4 = 26 - 11 * 2 \text{ gives}$$

$$1 = (26 - 11 * 2) * 3 - 11 = 26 * 3 - 11 * 7$$

$$\text{Thus } -7 * 11 \equiv 1 \pmod{26}, \text{ so } 11^{-1} \equiv -7 \equiv 19 \pmod{26}$$

$$11a \equiv 19 \pmod{26} \text{ by } 11^{-1} = 19$$

$$a \equiv 19 * 19 = 361$$

$$361 - 13 * 26 = 23 \pmod{26}$$

$$a = 23 \text{ } b = 14$$

$$y \equiv 23x + 14 \pmod{26}$$

14.

$$\phi(4) = 2 \text{ (1, 3)}$$

$$\phi(5) = 4 \text{ (1, 2, 3, 4)}$$

$$\phi(9) = 6 \text{ (1, 2, 4, 5, 7, 8)}$$

$$\phi(26) = 12 \text{ (1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25)}$$

15.1

Used frequency letter analysis to identify the most common letters r, o, c, and smart clues to find that RSC repeated a lot and GKNO was frequent. GKNO ended up mapping to Wade. Then slowly mapping each letter and figuring out the rest of the words when I could. Took a while.

15.2

on july 30th, 2014, 46-year-old wayne wade broke into a home in hollywood, florida. unluckily for wade, as he was casually loading a number of stolen valuables including a television and a coffee table into his pickup, the owner returned home and confronted him. flustered, he quickly jumped into his pickup and drove off, but in his understandable haste to depart the scene, he left his cell phone on the owner's bed. some time later, wade did what a lot of people who lost their smartphone would probably do: he called it in the hopes that someone would answer and return it. a police officer answered. wade told the officer

his name. he was arrested shortly thereafter. when wade appeared in court, judge john hurley was in awe of the burglar's stupidity. "so, the allegation is you burglarized the home, you left your cell phone and you told them what your name was over the cellphone," hurley said. "I'm just trying to absorb that." "because it was stolen, sir," wade explained.